

The Ephemeris

June 2014

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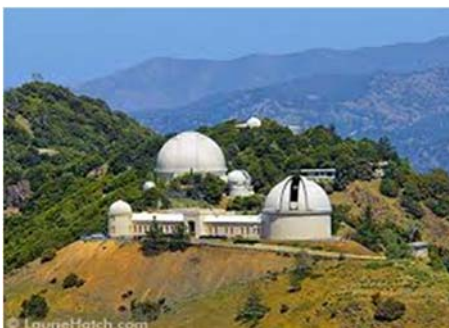
SJAA Matching Funds For Lick

From Michael Packer

Dear SJAA Members and Friends,

Some of you may have heard that Lick Observatory may close due to funding issues. Sandra Faber and other astronomers throughout California and beyond are working to save Lick for the simple reason that research there is now and in the future, viable; vital in fact.

Lick received \$1582 in donations from SJAA Members & Friends who attended Dr. Sandra Faber's speaking engagement on May 10th. Based on this preliminary number, SJAA will write a check to Lick for at least \$3082. From support of our Solar, Quick START and Advance Loaner program, Library and now the support the Lick Observatory, SJAA is proud to be a part of this community.



World Renowned Astrophysicist Dr. Sandra Faber, Director Of UC Observatories, speaking on The Research and Vision of Lick Observatory at Houge Park.

Help Save Lick! The observatory the South Bay sees from its own backyard.



June 2014 Events

Sunday, June 1

Solar observing (Houge): 2-4PM

Fix-It Day (Houge): 2-4PM

Friday, June 6

In-Town Star party (Houge): 9:30-11:30pm

Saturday, June 14

Houge Park

Board of Directors Meeting: 6-7:30pm

General Meeting: 7:30-10pm

Friday, June 20

Beginner Astronomy Class: 8:30-9:30pm

In-Town Star party (Houge): 9:45-11:45pm

Saturday, June 21

Starry Nights Star Party: 9:30 -11:30pm

Rancho Canada del Oro (RCDO)

Wed, June 25— Sun, June 29

Golden State Star Party

Saturday, June 28

Deep Sky Scavenger Hunt—details to follow

Mendoza Ranch

Sunday, July 6

Solar observing (Houge): 2-4PM

Fix-It Day (Houge): 2-4PM

SJAA events are subject to cancellation due to weather. Please visit website for up-to-date info.

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SJAA PROGRAM SPOTLIGHT— SIG

From Tom Piller

The Astro Imaging Special Interest Group (SIG), spearheaded by Harsh Kaushikkar, has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. The Introduction to Astro Photography presentation format is a collaborative effort between Harsh and John Wainwright.

As a part of the overall mission, several times a year Harsh speaks about different aspects of the Astro-imaging process. Most recently at the May 16th SJAA Imaging SIG presentation Harsh, with support from Mark Striebeck presented a short introduction to Astro-imaging which was attended by approximately 25 prospective and experienced imagers. The topics were very informative, at least from my beginner's perspective, and there were lots of questions and sharing of experience.

It's really great that Club members such as Harsh, Mark and John take the time to put together and present these programs to SJAA members and friends; I can't wait till the next one. In the mean time I have found that the best way to practice my beginner Astro-photographic skills is to get out to the dark sites, observe, ask lots of questions and of course, take pictures of the night sky.

SJAA SIG aspires to:

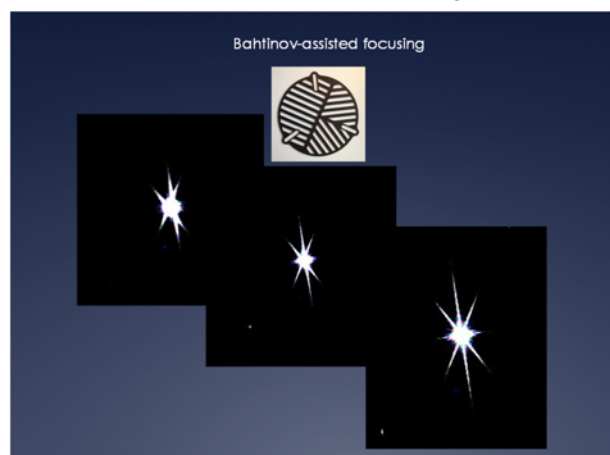
- be the place where we can physically meet and learn about the finer details of Astro-Imaging and techniques
- Provide a Physical Forum and Place to meet for Imagers and share their knowledge and experience

Photo: Heart Nebula Credit: Harsh-Kaushikkar



- v2**
- Capture
 - Scope OTA, DSLR, T-Ring, T-Ring adapter, Remote timer, EQ-mount
 - Focus
 - Laptop, Battery
 - Focus
 - Bahtinov mask
 - Guiding
 - Orion Mini Guider, PHD
 - Pre-processing and Stacking
 - Deep Sky Stacker
 - Registax
 - Post-processing
 - Photoshop

*Newtonian Imaging setup
Photo credit: John Wainwright*



- v3**
- Capture
 - Nebulosity
 - Focusing
 - Bahtinov mask
 - Guiding
 - PHD
 - Flats
 - Homebrew EL light box
 - Pre-processing
 - PixInsight
 - Post-processing
 - PixInsight
 - Photoshop

*Refractor Imaging setup
Photo credit: John Wainwright*

My Favorite Things, Vol. 1.3 Trek/There's More

by Teruo Utsumi

Star Trek

It's amazing what this 60s TV show did. It lit up the imagination of the public and inspired the generations since to pursue science and engineering. NASA recently presented an award to William Shatner.

Electromagnetic Spectrum Blindness

Sometimes there's more than "What you see is what you get."

For the past 400 years we've had great improvements in telescopes. Just a look at a few images taken by the Hubble Space Telescope will convince anybody that's true. And with that you might not even think to ask: "Is there more?" Imagine being colorblind and able to see only green. You'd be able to read this article (unless someone was really mean and set your computer to display any color but green), drive a car, and do any number of other things. Then if you gained full vision, you couldn't help but see all the things you couldn't perceive before.

Until the advent of electronics in recent decades, we were speaking electromagnetic (EM) spectrum blind — we could see what's in the sky through only a tiny part of the EM spectrum. Notice in the diagram visible below how narrow the range of visible light is compared to the entire spectrum and how much the atmosphere blocks.

We've had radio telescopes for decades and in just the last 10 to 20 years we've started to get good images in the infrared, ultraviolet, X-ray, and gamma-ray wavelengths. In other words we can "see" just about everything else now. Now try to imagine the excitement as astronomers experienced as they were first able to observe in each of these new EM regions.

Different EM regions give different clues. Like any good detective (Hardy Boys and Nancy Drew, anyone?), astronomers are putting these different clues together. One nice example of this is shown in a recent S&T [article](http://www.skyandtelescope.com/astronomy-news/supernova-remnant-in-technicolor); about a specific supernova. (www.skyandtelescope.com/astronomy-news/supernova-remnant-in-technicolor)

The first frame shows radio, then infrared, then visible, then X-ray, and finally all four. (Obviously, except for visible,

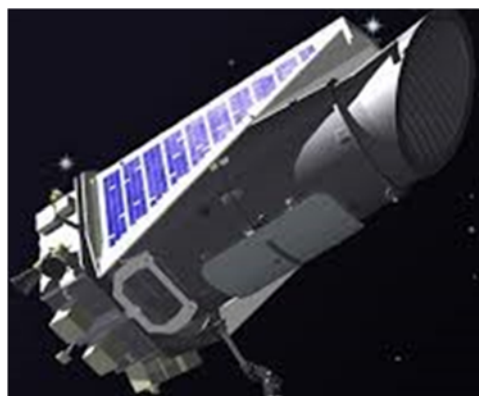
the images are shown in a false color since we can't directly see them.) Each shows the same region of space but different aspects of the supernova. Read the article for details.

This type of multispectral imaging has become more prevalent over the past several years, and this trend will only continue to grow.

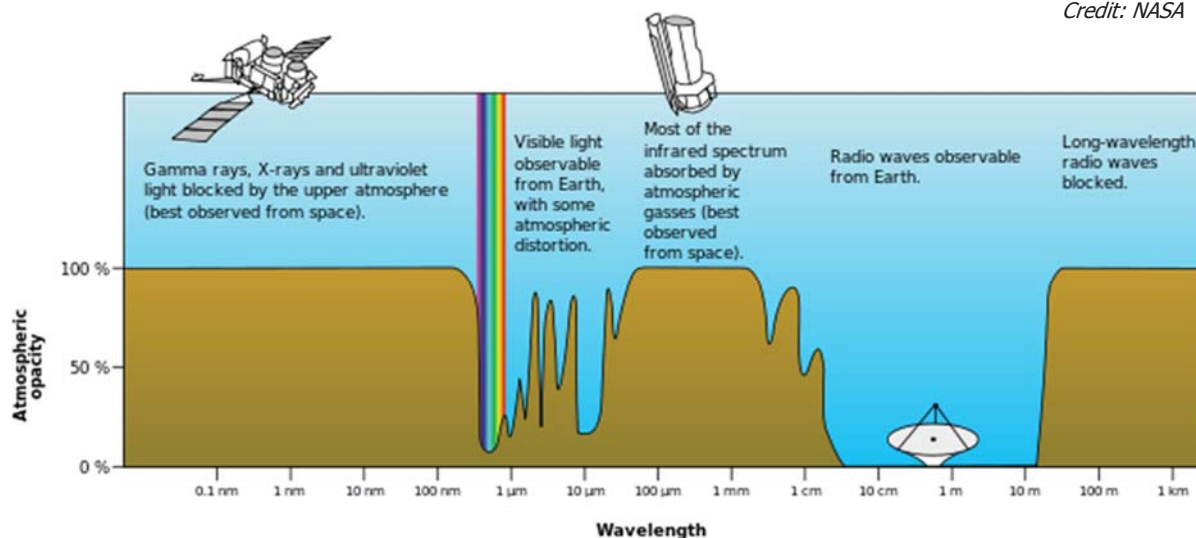
So what happens now that we've been cured of EMSB? (What an acronym!) Does this mean there are no other means to observe the universe? I'll let you think about it until next month. You'll see!

Editor's note: article to be continued in the July Ephemeris

Till next time, Clear Skies!



Kepler Space telescope
Credit: NASA



Graphic Credit: Wikipedia



Saturn is entering the early evening sky this spring just as Jupiter begins its exit in the west.

On May 10th, Saturn reached *opposition*, meaning it's opposite the Sun in the sky. It rises around sunset, glides across the sky all night, and sets near sunrise. Saturn is positioned among the dim stars of Libra, which keeps it relatively low in altitude even when it's highest in the middle of the night.

Cosmic Anniversary: 'Big Bang Echo' Discovered 50 Years Ago

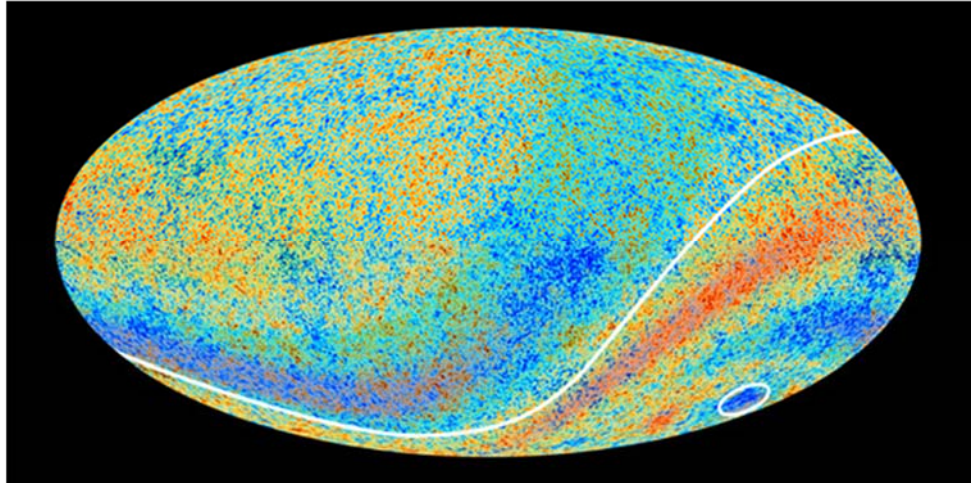
Humanity's understanding of the universe took a giant leap forward 50 years ago.

On May 20, 1964, American radio astronomers Robert Wilson and Arno Penzias discovered the cosmic microwave background (CMB), the ancient light that began saturating the universe 380,000 years after its creation. And they did so pretty much by accident.

Bell Labs' Holmdale Horn Antenna in New Jersey picked up an odd buzzing sound that came from all parts of the sky at all times. The noise puzzled Wilson and Penzias, who did their best to eliminate all possible sources of interference, even removing some pigeons that were nesting in the antenna.

"When we first heard that inexplicable 'hum,' we didn't understand its significance, and we never dreamed it would be connected to the origins of the universe," Penzias said in a statement. "It wasn't until we exhausted every possible explanation for the sound's origin that we realized we had stumbled upon something big."

And it was indeed big. Penzias and Wilson had spotted the CMB, the predicted thermal echo of the universe's explosive birth. The landmark find put the Big Bang theory on solid ground, suggesting that the cosmos did indeed grow from a tiny seed — a single point — about 13.8 billion years ago.

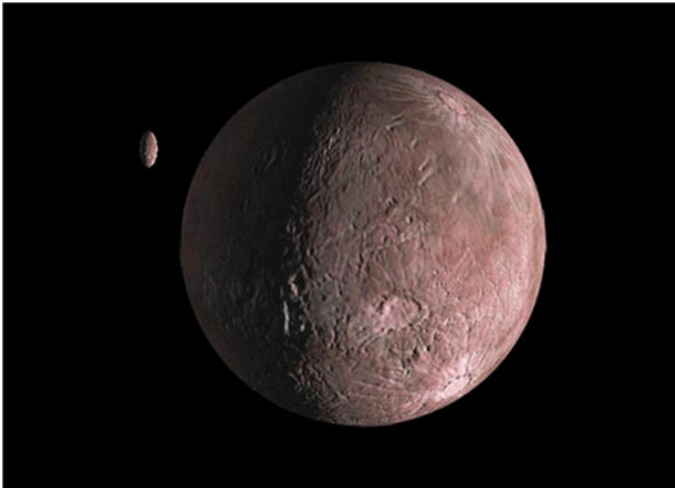


Two Cosmic Microwave Background anomalies hinted at by the Planck observatory's predecessor, NASA's WMAP, are confirmed in new high-precision data revealed on March 21, 2013. In this image, the two anomalous regions have been enhanced with red and blue shading to make them more clearly visible.

Credit: ESA and the Planck Collaboration

Credit: space.com

Quaoar: Planetoid Beyond Pluto



*An artist's conception of Quaoar and its small moon Weywot.
Credit: NASA/JPL-Caltech/R. Hurt (SSC-Caltech)*

Quaoar is a planetoid that lies beyond Pluto's orbit in the solar system. Its discovery in 2002, as well as subsequent discoveries of other small worlds, led to a new classification and the redefinition of Pluto as a "dwarf planet."

Quaoar is probably massive enough to be considered a dwarf planet, but it has not been classified as such yet. Quaoar lurks in the Kuiper Belt, a group of icy objects beyond Neptune. It is about 42 astronomical units, or Earth-sun distances, away. That's about 4 billion miles (6 billion km) — a billion kilometers more distant than Neptune. It takes about 288 years for Quaoar to go once around the sun in a roughly circular orbit.

Mike Brown and Chadwick Trujillo of the California Institute of Technology found Quaoar using a 48-inch telescope at the Palomar Observatory in California. It showed up as an 18.5-magnitude object moving against the background stars in the constellation Ophiuchus. They suggested the name Quaoar [pronounced *kwa-o-a*] after a creation god of the Native American Tongva tribe — the original inhabitants of the Los Angeles basin where Caltech is located. According to legend, Quaoar "came down from heaven; and, after reducing chaos to order, laid out the world on the back of seven giants. He then created the lower animals, and then mankind."

Credit: space.com

Mars Rover Curiosity Wraps Up Drilling Work, Prepares for Long Drive

NASA's Mars rover Curiosity is about to hit the road again.

The 1-ton Curiosity rover has finished its drilling operations at a sandstone slab dubbed "Windjana," successfully delivering collected samples to its onboard scientific instruments for analysis. The robot will soon resume the long drive toward the base of Mount Sharp, which has long been its ultimate destination, mission officials said.



NASA's Curiosity Mars rover used the Mars Hand Lens Imager (MAHLI) instrument on its robotic arm to illuminate and record this nighttime view of the sandstone rock target "Windjana." Image released May 15, 2014.

Credit: NASA/JPL-Caltech/MSSS

The rover bored a 2.5-inch-deep (6.4 centimeters) hole into Windjana earlier this month, performing its third full sample-collecting drilling activity since touching down inside Mars' huge Gale Crater in August 2012.

The previous two drilling operations, which investigated fine-grained mudstones at a site called Yellowknife Bay, allowed mission scientists to determine that the Red Planet could have supported microbial life billions of years ago.

Yellowknife Bay is close to Curiosity's landing site. Windjana is about 2.5 miles (4 kilometers) southwest, on the way toward the foothills of Mount Sharp, which rises more than 3 miles (5 km) into the Martian sky from Gale Crater's center.

Mission scientists want Curiosity to climb up through these foothills, reading a history of the Red Planet's changing environmental conditions as it goes. The rover's measurements could help shed light on why and



This view from the Mars Hand Lens Imager (MAHLI) on NASA's Curiosity Mars Rover shows the rock target "Windjana" and its immediate surroundings after inspection of the site by the rover. The drilling of a test hole and a sample collection hole produced the mounds of drill cuttings that are less red than the other visible surfaces. Image released May 15, 2014.

Credit: NASA/JPL-Caltech/MSSS

when Mars changed from a relatively warm and wet planet to the cold and dry world we know today, NASA officials have said.

Curiosity is exploring the Red Planet with the aid of 10 different scientific instruments. Two of the workhorses are the Chemistry and Mineralogy instrument (CheMin) and the Sample Analysis at Mars instrument (SAM), both of which sit on the rover's main body.

CheMin and SAM will perform a detailed analysis of the Windjana samples, as they did for the powder collected at Yellowknife Bay.

While Windjana powder has already been delivered to both CheMin and SAM, Curiosity carries still more of the material, which may be transferred to the instruments during pauses in the rover's driving if need be, NASA officials said.

Credit: space.com



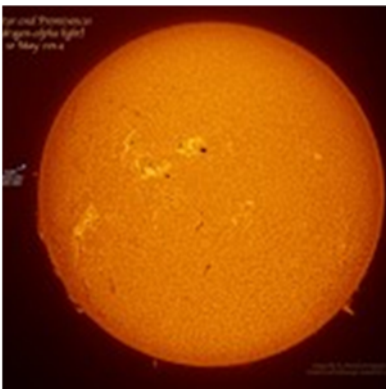
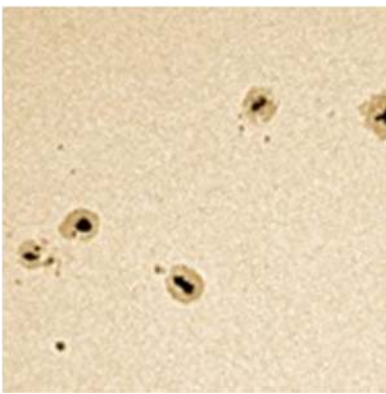
Astronomy Day at MLK Library May 10, 2014

From Michael Packer

With A Sunspot number of 123 (NOAA) and several active regions, notably AR2055, 2056 and 2057, El Sol was a plasma rock on a roll during National Astronomy Day 2014.

The H-Alpha Flares – solar prominences – were also putting on a great show. All in all we shared astronomy with about 75 patrons of the library and SJSU Campus.

Many thanks to MLK Library support from Ashour Benjamin, Reed Duong, and Deborah Estreicher, along with SJAA Volunteers Marion Barker, Paul Colby, Terry Kahl, Kevin Lahey, Bill O'Neil, Michael Packer, and Teruo Utsumi.



*Sun Images
Photo credit Robert Duvall*



Kid Spot Jokes:

1. **What is an astronaut's favorite snack?**
Space Chips.
2. **Why did the astronaut bring paper and paint on his trip?**
So he could do space crafts.

Kid Spot Quiz:

1. **When was Hubble launched?**
2. **What is Ganymede?**

Kid Spot Night Sky Challenge: June 2014

See if you can spot the following objects in the sky:

- ⇒ Saturn – Evening sky
- ⇒ Titan — Saturn's bright moon
- ⇒ M13 Globular Cluster in Hercules
- ⇒ Double Double in Lyra
- ⇒ Mars—one hour after sunset

<http://skyandtelescope.com/observing/ataglance>



Image of Saturn
Photo Credit: Wikipedia

Constellations

Hercules - is a constellation named after the Roman mythological hero adapted from the Greek hero Heracles. Hercules was one of the 48 constellations listed by the 2nd century astronomer Ptolemy, and it remains one of the 88 modern constellations today. It is the fifth largest of the modern constellations.

Hercules contains two bright globular clusters: M13, the brightest globular cluster in the northern hemisphere, and M92. It also contains the nearly spherical planetary nebula Abell 39. M13 lies between the stars η Her and ζ Her; it is dim, but may be detected by the unaided eye on a very clear night.

M13, visible to both the naked eye and binoculars, is a globular cluster of the 6th magnitude that contains more than 300,000 stars and is 25,200 light-years from Earth. It is also very large, with an apparent diameter of over 0.25 degrees, half the size of the full moon; its physical diameter is more than 100 light-years. Individual stars in M13 are

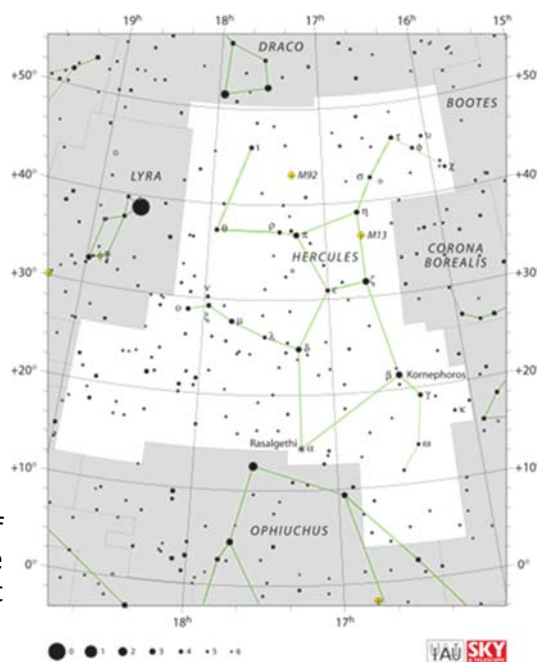
resolvable in a small amateur telescope.

M92 is a globular cluster of magnitude 6.4, 26,000 light-years from earth. It is a Shapley class IV cluster, indicating that it is quite concentrated at the center; it has a very clear nucleus. M92 is visible as a fuzzy star in binoculars, like M13; it is denser and smaller than the more celebrated cluster. The oldest globular cluster known at 14 billion years, its stars are resolvable in a medium-aperture amateur telescope.

The traditional visualization imagines a Hercules as Hercules's head; its name, Ras Algethi, literally means "head of the kneeling one".

Hercules is also sometimes associated with Gilgamesh, a Sumerian mythological hero.

Source: Wikipedia



Credit: IAU / Sky & Telescope

Kid Spot Quiz Answers:

- 1) April 24, 1990
- 2) One of Jupiter's moons

Programs

School Events

From Jim Van Nuland

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill. Check out the instructions for setting up your school star party on the SJAA web page under the Classes/Programs tab. Jim Van Nuland is the coordinator.

Public Star Parties

Check the event schedule on the SJAA web page for more information.



Photo: Scopes setting up at the May 23rd Houge Park Star Party.

Beginners Astronomy Class

Check the event schedule on the SJAA web page or June 2014 events on Page 1 for more information. The classes are held in conjunction with one of the monthly Houge Park Star Parties. The most recent class was presented by Rich Neuschaefer on May 23rd. Approximately 50 students from Rolling Hills Middle School attended the class and star party.

Advanced Loaner Telescopes

From Dave Ittner



Photo: Dave Ittner familiarizing new SJAA member with the Club's motorized 8" dobsonian for the Club Loaner Program.

The purpose of the loaner program is for SJAA members to evaluate equipment they are considering purchasing or are just curious about. Check out the growing list of equipment on the web page under 'Programs'. Please note that certain items have restrictions or special conditions that must be met.

Fix-It

From Ed Wong

If you plan on bringing something to the FixIt session inside Building 1, though it's not required, we ask that you check out the FixIt page on the SJAA website and fill out the form so we know what to expect. Here's where to find it all:

<http://www.sjaa.net/fixit>

Astro Imaging Special Interest Group (SIG)

The Imaging SIG meets roughly every other month at Houge Park to discuss topics about imaging, as well as in the field, usually at Rancho Cañada del Oro. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

See program spotlight article on page 2.

Club Library

The Club Library is now up and running. Contact Club Librarian Sukhada Palav for additional information.

From the Board of Directors

General Notices

The Golden State Star Party (GSSP) will take place June 25-29, 2014. See Michael Packer or the SJAA blog for more details.

The annual SJAA Yosemite star party will be held on July 25 and 26, 2014, at Glacier Point in Yosemite National Park. See the SJAA webpage for more information.

There is currently one (1) board seat open that needs to be filled. See Lee Hoglan or any board member if you are interested.

Board Meeting Excerpts May 10, 2014

In attendance

Lee Hoglan, Ed Wong, Dave Ittner, Teruo Utsumi, Rich Neuschaefer, Michael Packer

Scavenger Hunt

Ed Wong reported that the scavenger hunt is scheduled for weekend of June 27th at Mendoza Ranch. The event is targeted at beginners and the initial plan is to have a set of 50 objects although it may be pared down to 30 to make it easier for beginners to finish.

QuickSTART request

The Board approved Dave Ittner's request to purchase 8 copies of the Sky & Telescope Pocket Sky Atlas to be used in conjunction with the Quick START loaner program.

Bear Creek Stables

At the event there was a good scope turnout and approximately 40 people attended. Follow-up discussion at the Board meeting covered what worked, what didn't work, developing a relationship with Bear Creek Stables and future use. The current plan is to use the facility approximately 2 times per year.

SJAA Contacts

President:	Rob Jaworski
Vice President:	Lee Hoglan
Treasurer:	Michael Packer
Secretary:	Teruo Utsumi
Director:	Greg Claytor
Director:	Dave Ittner
Director:	Ed Wong
Director:	Rich Neuschaefer
Director:	pending
Beginner Class:	pending
Fix-it Program:	Ed Wong
Imaging SIG:	Harsh Kaushikkar
Library:	Sukhada Palav
Loaner Program:	Dave Ittner
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Production	Tom Piller
Publicity:	Rob Jaworski
Questions:	Lee Hoglan
Quick START	Dave Ittner
Solar:	Michael Packer
School Events:	Jim Van Nuland
Speakers:	Teruo Utsumi

E-mails: <http://www.sjaa.net/contact-us/>

SJAA Ephemeris newsletter of the San Jose Astronomical Association, is published monthly

Articles for publication should be submitted by the 20th of the previous month.

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Membership Type:

☐ Regular — \$20

☐ Regular with Sky & Telescope — \$53

☐ Junior (under 18) — \$10

☐ Junior with Sky & Telescope — \$43

<http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to
sjaamemberships@gmail.com

Subscribing to Sky & Telescope magazine through the SJAA saves you \$5 off the regular rate. (S&T will not accept multi-year subscriptions through the club program. Allow 2 months lead time.)

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at <http://www.sjaa.net/join-the-sjaa/>

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